RAJALAKSHMI ENGINEERING COLLEGE

V R Krithika Sree 250701371-2025-CSE V2

Started on	Wednesday, 16 September 2026, 8:40 AM
State	Finished
Completed on	Wednesday, 16 September 2026, 8:47 AM
Time taken	6 mins 51 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int found=0;
void bSearch(int arr[], int l, int h, int key){
    if(l>=h){
        return;
    }
    int sum = arr[l]+arr[h];
    if(sum==key){
        printf("%d\n", arr[l]);
        printf("%d\n", arr[h]);
        found =1;
        return;
    }
    else if(sum<key){
        bSearch(arr, l+1,h, key);
    }
    else{
        bSearch(arr,l,h-1, key);
    }
}

int main(){
    int n;
    scanf("%d", &n);
    int arr[n];
    for(int i=0; i<n; i++)
        scanf("%d", &arr[i]);
    int x;
    scanf("%d", &x);
    bSearch(arr,0,n-1,x);
    if(!found){
        printf("No\n");
    }
    return 0;
}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔

	Input	Expected	Got	
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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